Same Pizza, Different Slices?

Putting the No-Tax Assumption of Modigliani–Miller Proposition I on Trial

Lao Xiusheng, Xu Riqi, Wu Kaluk, Jiang Yumeng, Li Jiabao and Li Jingwen
Group D.E. Show&Tell, MH6806 Principles of Finance and Risk Management (Topic 3)
Nanyang Technological University, October 2026

Abstract

Modigliani and Miller's Proposition I states that, in frictionless markets, the value of a firm is independent of its capital structure. We put a single assumption on trial, the absence of corporate income tax, and ask whether the interest tax shield turns debt into a free lunch. In a one-period, two-state economy, debt raises firm value by exactly the present value of the tax shield, but only up to the largest debt the weak state can service; beyond that point distress costs dominate. With a continuous cash-flow distribution the value function has an interior optimum, although the gain is small, under 1% of unlevered value. What surprised us is that a higher tax rate makes each unit of debt more valuable but lowers the amount of debt that is safe, because the tax also comes out of the weak state's cash. We then ask whether this small gain reflects the mechanism itself or the simplifications of the setting. Letting lenders price default risk, limiting the usability of interest deductions and allowing debt to be rolled over all leave the structure of the trade-off intact; personal taxes can shrink the advantage or remove it, and the size of the gain depends on the horizon: with perpetual debt it rises to roughly 10% of firm value. Applications to the 2017 US Tax Cuts and Jobs Act, the Singapore tax system and two Singapore-listed firms show that the model ranks debt capacity as the trade-off theory predicts, although its "no debt" verdict for the airline is driven by our no-cash-buffer default rule, and that observed leverage also reflects frictions outside the model. We conclude that the corporate tax makes financing matter without making debt free.

Keywords: capital structure; Modigliani–Miller; interest tax shield; financial distress; trade-off theory
JEL classification: G32, G33, H25

1 Introduction

Our starting point is the two-state firm of Lecture 1 (pp. 9, 47–48):[1] financed by equity alone or by debt and equity, it pays the same 120 or 60, so buying the cheaper package and shorting the dearer one would be a money machine whenever their prices differed. Modigliani and Miller (1958) turn this argument into the benchmark result: when capital markets are perfect and the firm's investment policy is held fixed, financing decisions cannot change firm value:

$$V_{\mathrm{L}} = V_{\mathrm{U}}. \tag{1}$$

Debt and equity merely partition the same operating cash flows; to borrow a familiar image, the size of the pizza does not depend on how it is sliced. The proposition is valuable less as a description of the world than as a benchmark: if capital structure affects value in practice, at least one of its assumptions must fail, and the useful question becomes which one, and by how much.

We study the assumption with the most direct consequences for practice: the absence of corporate taxation. Because interest is deductible in almost every corporate tax system, debt reduces the government's claim on operating income, and Modigliani and Miller (1963) show that this alone makes financing matter. Corporate taxation is also the friction that policy changes most directly. The 2017 US Tax Cuts and Jobs Act lowered the federal rate from 35% to 21% and capped interest deductions, while Singapore combines a 17% corporate rate with broad exemptions for investment income at the investor level. Lecture 1 (p. 50) lists tax shields among the benefits of debt and financial distress among its costs; we put numbers on that trade-off. Our research question is therefore: *does the corporate income tax turn debt into a free lunch?*

We judge every result against two conditions. (i) *Attribution*: any value created by debt must vanish as the tax rate goes to zero, so that it can be attributed to the tax. Inside our model this holds by construction, because tax is the only benefit of debt; outside it, S-REITs, which pay almost no tax but still borrow about 40% of their assets, show that leverage alone is not evidence of a tax benefit (Section 6.2). (ii) *Robustness*: the shape of the result, a gain up to a debt limit followed by a loss, must survive reasonable changes in parameters; findings that depend on a particular parameter value are reported as properties of the model rather than of the world.

Our analysis draws on a well-established literature. Modigliani and Miller (1963) show that with corporate taxes $V_{\mathrm{L}} = V_{\mathrm{U}} + \tau_{\mathrm{c}} D$, which taken literally implies all-debt financing. The trade-off theory bounds this result with bankruptcy costs (Kraus and Litzenberger, 1973). Miller (1977) shows that personal taxes on interest income

[1] "Lecture" refers to the MH6806 course slides (Vaidya, 2026); page numbers are those of the PDF files.

can offset the corporate advantage, and DeAngelo and Masulis (1980) show that the marginal value of interest deductions falls when they cannot always be used. Empirically, Graham (2000) estimates the capitalised tax benefit of debt at 9.7% of firm value and documents that many firms borrow conservatively, while distress costs are estimated at 10–20% of firm value once distress occurs (Andrade and Kaplan, 1998) and, in risk-adjusted present value, at about 4.5% for a BBB-rated firm (Almeida and Philippon, 2007).

Our answer is no: the corporate tax makes debt valuable to shareholders, but the gain is paid for by the government and capped by the firm's weakest year, so it is not free. We follow this mechanism from the two-state firm to calibrated settings. Sections 2–4 present the base model and test its robustness. Section 3 also reruns the law-of-one-price trade with taxes and reads the result with CAPM betas, WACC and a stress test (Lectures 1, 2 and 5). Section 5 relaxes, one at a time, the four simplifications most likely to overturn the result. Section 6 confronts the model with the 2017 US tax reform, the Singapore tax system and the reported cash flows of two Singapore-listed firms. Section 7 concludes, and Table 5 summarises which results survive.

2 Model

We use the two-state firm of Lecture 1 (pp. 9, 47) in a one-period economy. Operating cash flow (EBIT) next year is $X_s \in \{120, 60\}$ in the strong and weak state, each with probability 0.5; the firm then liquidates. All claims are valued at their expected payoff discounted at $r = 10\%$. We treat r as the risk-free rate and 0.5/0.5 as risk-neutral probabilities, which is why debt, equity and the tax shield can all be discounted at 10% even though equity is riskier (Section 3.4). In the benchmark there are no frictions. The assumption on trial is the absence of corporate tax: the firm pays corporate tax at rate $\tau_c = 20\%$ on EBIT less interest, interest is deductible, also in a default year, but principal is not, and there is no loss carry-forward.

The firm borrows D at date 0 at a fixed rate $r_D = 10\%$ and promises $F = (1 + r_D)D$ at date 1. After-tax cash in state s is $A_s = X_s - \tau_c \max(X_s - r_D D, 0)$. If $A_s < F$ the firm defaults and a fraction $c = 20\%$ of A_s is lost to distress (legal costs, fire sales, lost customers). Distress costs are a supporting assumption, not a second defendant: they only explain why a taxed firm would not borrow without limit, and they never create value. Firm values are

$$V_U = \frac{\mathrm{E}[(1-\tau_c)X]}{1+r}, \qquad V_L = \frac{\mathrm{E}[A - cA \cdot \mathbf{1}\{A < F\}]}{1+r}. \tag{2}$$

Because the cash flows, the investment policy and the discount rate never change, any change in value is attributable to financing. All parameter values are illustrative; Table A1 lists them and Appendix B gives the derivations.

3 Results in the base model

3.1 *The no-tax benchmark*

Without taxes the all-equity firm U is worth $90/1.1 = 81.8$. A levered firm that promises creditors 80 pays them 80 in the strong state and the entire 60 in the weak state, where it defaults; shareholders receive 40 or 0. Debt is worth $70/1.1 = 63.6$ and equity $20/1.1 = 18.2$, which sum to 81.8. Because debt plus equity pays exactly what firm U pays in every state, the law of one price implies (1): if the two packages were priced differently, buying the cheaper one and shorting the dearer one would lock in a profit today with zero payoff in both states, the trade of Lecture 1 (pp. 48, 60). Investors could reach the same result by homemade leverage, borrowing on personal account to turn U's shares into L's equity. Both routes assume no taxes, distress or trading costs (Lecture 1, p. 48).

3.2 *Corporate tax and the interest tax shield*

With a 20% tax the unlevered firm pays 24 or 12, so $V_U = 72/1.1 = 65.45$ and the present value of the government's claim is 16.36. A firm with $D = 40$ deducts interest of 4, so taxable income is 116 or 56 and tax is 23.2 or 11.2. Its investors receive 0.8 more in each state, and

$$V_L - V_U = \frac{\tau_c r_D D}{1+r} = \frac{0.20 \times 0.10 \times 40}{1.10} = 0.73, \tag{3}$$

so that $V_L = 66.18$. The pie itself is unchanged, since V_L plus the present value of taxes (15.64) still equals 81.8. Debt creates no cash flow; it transfers 0.73 from the government's claim to investors. Since (3) is proportional to τ_c, the gain vanishes at $\tau_c = 0$ and the attribution condition is satisfied.

3.3 *Debt capacity and financial distress*

If each unit of debt added value, the firm would borrow without limit. It cannot, because the weak state's after-tax cash, $60 - 0.2(60 - 0.1D) = 48 + 0.02D$, must cover the promised repayment $1.1D$. The two are equal at

$$D^* = \frac{(1-\tau_c)X_w}{(1+r_D) - \tau_c r_D} = \frac{48}{1.08} = 44.4. \tag{4}$$

Below D^* every unit of debt adds shield (Table 1). At $D = 45$ the weak state defaults and loses 20% of its 48.9 of cash (9.78), and $V_L - V_U$ falls from +0.80 to −3.63 (Figure 1a).

Table 1. Debt and firm value in the two-state model ($\tau_c = 20\%$, $c = 20\%$)

Debt D	Weak-state after-tax cash	Promised repayment	Weak state defaults	$V_L - V_U$
40	48.80	44.00	No	0.73
44	48.88	48.40	No	0.80
45	48.90	49.50	Yes	−3.63
50	49.00	55.00	Yes	−3.55

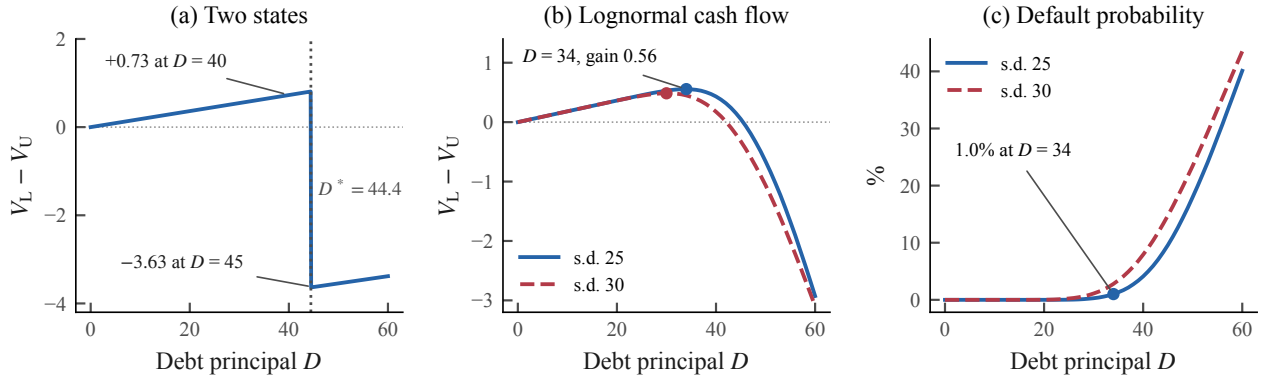


Figure 1. Value of debt financing. (a) Two-state model: the gain rises to 0.81 at $D^* = 44.4$ and then collapses. (b) Lognormal cash flow with mean 90 and standard deviation 25 (200,000 draws), all other assumptions unchanged: the value function has an interior optimum near $D = 34$. (c) Default probability in the lognormal model. Dashed lines in (b) and (c) use a standard deviation of 30.

Where the money machine breaks down. Rerun the trade of Lecture 1 (pp. 48, 60) with taxes: buy all of L's debt and equity and short U. At $D = 40$ it pays +0.8 in both states, so if the market priced L at $V_U = 65.45$ it would be a riskless money machine whose counterparty is the tax authority. Homemade leverage cannot copy it, because interest paid by an investor does not reduce the firm's tax bill; the law of one price survives and prices the shield, bidding V_L up to $V_U + 0.8/1.1 = 66.18$, which is (3). Borrowing more to collect more shield stops at $D^* = 44.4$: at $D = 45$ the weak state defaults and loses 9.78 of its 48.9 of cash, and the position pays +0.90 in the strong state but −8.88 in the weak one, worth −3.63. Without distress costs it would stay riskless beyond D^* (Section 4.1), so it is costly default, not the debt limit itself, that switches the machine off.

3.4 Reading the result with the course toolkit

Three tools from the lectures read the base case directly. First, leverage moves risk between the slices while the pizza is unchanged (Lecture 2, p. 5). With $D = 40$, equity pays 52.8 or 4.8 and is worth 26.18; its payoff swings by 48, exactly as much as U's (96 or 48), but on a base of 26.18 instead of 65.45. Because debt and the one-year shield are riskless, all market risk sits in equity, so $\beta_E = \beta_U \times V_U/E = 2.5\beta_U$ and, under CAPM, equity's risk premium is 2.5 times the unlevered firm's: a one-period Proposition II with taxes. Hamada's $1 + (1 - \tau_c)D/E = 2.22$ is lower because it assumes a perpetual shield worth $\tau_c D = 8$ rather than 0.73. Second, the WACC of Lecture 1 (p. 52) carries the same shield: $(26.18 \times 10\% + 40 \times 10\% \times 0.8)/66.18 = 8.79\%$, and discounting U's after-tax 72 at 8.79% returns exactly $V_L = 66.18$. Third, the debt limit is a stress test in the sense of Lecture 5 (p. 23): (4) asks whether creditors are paid in the weak state and needs no probability. The lognormal optimum has a VaR reading: at $D = 34$ the promise of 37.4 sits at the 1% quantile of after-tax cash (37.4). The level is not a law; with $c = 5\%$ or 40% the optimum accepts about 5% or 0.4% default risk. The tax rate sets the prize; the cost of failure, like expected shortfall beyond VaR (Lecture 5, pp. 19–20), sets how much tail risk the firm should accept.

4 Robustness

4.1 Parameter sensitivity

Figure 2 varies one parameter at a time. A higher tax rate raises the shield per unit of debt but also reduces weak-state cash, so the debt limit falls: $D^* = 46.0$, 44.4 and 39.3 at $\tau_c = 17\%$, 20% and 30%. A wider spread between the states, holding mean EBIT at 90, has the same effect ($D^* = 51.9$, 44.4 and 37.0). The value 44.4 is

therefore a property of our parameters rather than a general rule. In every case, however, the limit is set by what the weak state can repay, and the gain below it is small, between 1.0% and 1.9% of V_U.

The distress cost does not move D^*; it determines how far value falls beyond it (−0.3, −3.6 and −8.1 at $D = 45$ for $c = 5\%$, 20% and 40%). There is one qualification. If c is below about 3.6%, the additional shield outweighs the distress cost even beyond the limit, so debt just above D^* still adds value; at $c = 5\%$ the gain turns negative at D^* but becomes positive again for D above 61.5, because the defaulting weak state still deducts its full interest, which it never pays (Section 2), an artefact we do not interpret. The conclusion that debt beyond the limit destroys value therefore requires distress costs that are not negligible, consistent with the empirical estimates cited in Section 1.

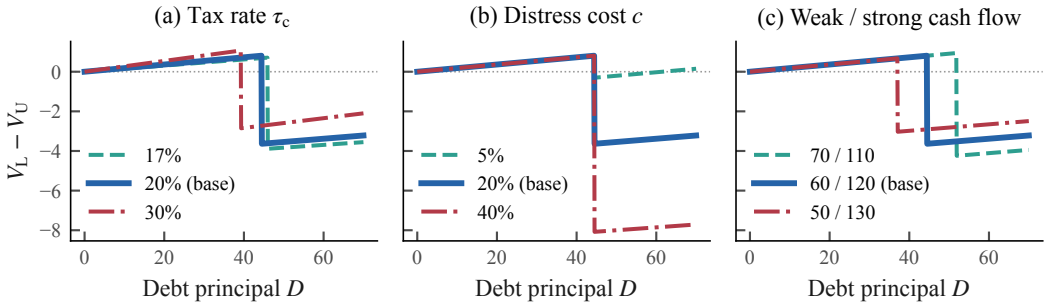


Figure 2. Sensitivity of the two-state model to one parameter at a time. The thick solid line is the base case. In panel (c) mean EBIT is held at 90.

4.2 Continuous cash flows

The vertical drop in Figure 1a is an artefact of having only two states. With a lognormal cash flow of the same mean (standard deviation 25) and every other assumption unchanged, the value function becomes smooth and concave, the familiar trade-off between tax shield and expected distress cost (Kraus and Litzenberger, 1973). The optimum is near $D = 34$, where the default probability is about 1% and the gain is 0.56, or 0.85% of V_U; the gain returns to zero at $D \approx 45.5$ (Figure 1b–c). With a standard deviation of 30, which matches the dispersion of the two-state model, the optimum is near $D = 30$ with a gain of 0.49.

Both conditions are met. With $\tau_c = 0$ the gain is exactly zero up to the debt limit in the two-state model and never positive in the lognormal model, and every specification produces a positive gain followed by a limit, provided distress destroys more than a few per cent of default-state cash (3.6% in the base case).

5 Extensions

The base model rests on four simplifications that could overturn its conclusion. Each extension below relaxes one of them; Appendix C reports the detailed results.

5.1 Market-priced debt

The base model holds the borrowing rate at 10% regardless of leverage, so lenders implicitly accept losses once default becomes possible. We now require lenders to break even: the promised repayment F solves $D = \mathrm{E}[\text{payoff to creditors}]/(1 + r)$, and the full coupon $F - D$ is deductible. In the lognormal model the break-even rate is 10.4% at the optimum ($D \approx 35$), 13.8% at $D = 45$, 19.3% at $D = 50$ and 32.5% at $D = 55$ (Figure C1b; Table C1 gives the two-state case, Table C2 the lognormal). The optimal debt level and the maximum gain are almost unchanged ($D = 35$ and 0.57, against 34 and 0.56).

What changes is the allocation of distress costs (Figure C1a). Under a fixed rate, shareholders gain $V_L - V_U$ plus the lenders' loss, a quantity that rises with debt (2.1 at $D = 50$ and 5.1 at $D = 60$), so they would choose leverage far above the value-maximising level. With fair pricing, expected distress costs are borne by shareholders through the interest rate, and maximising shareholder wealth coincides with maximising V_L. The fixed-rate assumption is thus innocuous for firm value but distorts incentives, which is why lenders rely on covenants, ratings and credit spreads. This is the agency conflict that Lecture 1 (p. 50) lists among the costs of debt: at $D = 50$, shareholders who borrow at a rate that ignores default risk gain 2.06 while the firm as a whole loses 0.66 (Table C2).

5.2 Usability of the interest deduction

Because the base firm always earns at least 60, every unit of interest reduces tax. Actual firms have loss years, and many tax codes cap interest deductions; the United States, for example, limits net interest to 30% of adjusted taxable income (Section 6.1). We draw EBIT from a normal distribution with a given volatility (standard deviation

over mean) and compute the share of $\tau_c \times$ interest that is actually saved as a function of interest coverage, E[EBIT]/interest (Figure 3a, Table C3). Our firm, with volatility 0.28 and coverage of about 20 at $D = 44$, uses essentially all of its shield. A cyclical firm with volatility 1.0 loses about 17% of its shield even at a coverage of 10, because of loss years. A 30%-of-EBIT cap binds in an average year once coverage falls below 3.3 (and in bad years even above it); at a coverage of 2 only about 60% of the shield is usable. The effective tax rate on interest is therefore below the statutory rate, and most so when leverage is high, which adds a tax-side limit to debt alongside distress costs (DeAngelo and Masulis, 1980).

5.3 *Personal taxes*

If lenders pay tax τ_D on interest and shareholders pay τ_E on equity income, the net tax advantage of one unit of interest is (Miller, 1977)

$$T^* = 1 - \frac{(1-\tau_c)(1-\tau_E)}{1-\tau_D}, \tag{5}$$

which equals τ_c only when $\tau_D = \tau_E$. Table 2 shows that investor-level taxes can matter as much as the corporate rate. For a top-bracket US individual the net advantage was 12.5% before the 2017 reform and is slightly negative afterwards, whereas for a tax-exempt investor it simply equals τ_c. Whether debt offers any tax advantage at all therefore depends on the identity of the marginal investor.

Table 2. Net tax advantage of debt per unit of interest, T^*

Setting	τ_c	τ_D (interest)	τ_E (equity)	T^*
Base model (no personal taxes)	20%	0%	0%	20.0%
United States 2017, top-bracket individual	35%	43.4%	23.8%	12.5%
United States from 2018, top-bracket individual	21%	40.8%	23.8%	−1.7%
United States, tax-exempt investor (2017 → 2018)	35% → 21%	0%	0%	35% → 21%
Singapore, individual investor	17%	0%	0%	17.0%
S-REIT distributing its income	≈ 0%	0%	0%	≈ 0%

Notes: US rates are top federal statutory rates including the 3.8% net investment income tax; deferral of capital gains would lower the effective τ_E further. Sources are given in Section 6.

5.4 *Multi-period debt*

One-period debt generates a shield for one year only. If the same debt is maintained for T years, the shield is worth

$$\mathrm{PV(shield)} = \tau_c D\left[1 - (1+r)^{-T}\right], \tag{6}$$

which equals 0.73 for one year, 3.03 for five, 4.92 for ten and $\tau_c D = 8.0$ for perpetual debt (Table C4), the result of Modigliani and Miller (1963). To compare like with like, we model a perpetual firm with the same annual EBIT distribution ($V_U = 720$) that issues perpetual debt and defaults in the first year in which EBIT falls short of interest, losing a fraction c of its value. With annual default probability p, Appendix B shows that

$$V_L - V_U = \frac{(1-p)\tau_c r D - pcV_U}{r+p}. \tag{7}$$

The trade-off survives, but the gain is an order of magnitude larger (Figure 3b): the optimum is at $D/V_U = 57\%$, with interest coverage of 2.2 and an annual default probability of 0.3%, and it adds 10.4% to firm value. Doubling EBIT volatility lowers optimal leverage to 27% and the gain to 4.5%. The one-period horizon therefore understates the magnitude of the gain. For comparison, Graham (2000) estimates the gross tax benefit of debt at 9.7% of firm value (4.3% net of personal taxes); in our model the gross shield at the optimum is 11.1%, of which 10.4% remains after expected distress costs.

6 Applications

6.1 *The 2017 US Tax Cuts and Jobs Act*

The Tax Cuts and Jobs Act (TCJA) changed precisely the assumption under examination. It reduced the federal corporate rate from 35% to 21% for tax years beginning after 2017 and, through Section 163(j), limited net business interest deductions to 30% of adjusted taxable income (temporarily 50% for 2019–2020 under the CARES Act), defined approximately as EBITDA in 2018–2021 and as EBIT in 2022–2024, with disallowed interest carried forward (Gale et al., 2018; IRS, 2025, 2026). The One Big Beautiful Bill Act of 2025 restored the EBITDA-based definition for tax years beginning after 31 December 2024 (IRS, 2026); the cut in the

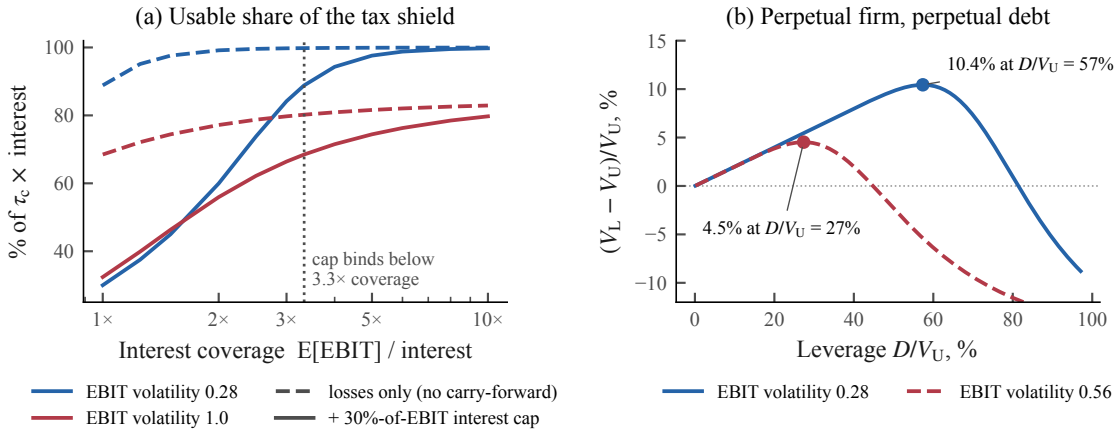


Figure 3. (a) Share of the statutory tax shield that is used, by interest coverage, for normally distributed EBIT with volatility (standard deviation over mean) of 0.28 or 1.0. Dashed lines allow for loss years only; solid lines add a 30%-of-EBIT interest cap. (b) Perpetual firm with perpetual debt: value gain against leverage for lognormal EBIT with volatility 0.28 and 0.56.

top individual rate on interest income from 39.6% to 37% came from the TCJA itself in 2018 and was made permanent by the 2025 Act.

Table 3 introduces these changes into the multi-period model of Section 5.4 and yields three predictions. First, the rate cut reduces the value that debt adds by about 40% (from 137 to 79 for the stable firm, or from 23.4% to 11.1% of a V_U that itself rises from 585 to 711) but has a much smaller effect on how much a firm should borrow: optimal debt falls by about 3% (13% for the volatile firm), because the optimum is determined more by how quickly default risk rises in bad years than by the tax rate. Second, the interest cap is what reduces borrowing: on its own it lowers optimal debt by about a third and raises optimal interest coverage from 2.1 to 3.1, because interest above 30% of EBIT earns no shield and the extra tax makes default more likely (Section 5.2). Since our cap is based on EBIT, the stricter 2022–2024 rule, and disallowed interest is lost rather than carried forward, a reduction of one third should be read as an upper bound. Third, at the investor level, the net advantage for a top-bracket individual fell from 12.5% to slightly below zero (−1.7%) (Table 2).

The empirical evidence is broadly consistent with these predictions, though not unanimous. Carrizosa, Gaertner and Lynch (2023) find that firms constrained by the interest limitation reduced leverage by 7.6% of assets relative to unconstrained firms, mainly by issuing less new long-term debt. Using a regression discontinuity around the $25 million small-business exemption, Sanati and Beyhaghi (2026) find that corporate debt declines significantly as the tax benefits of debt shrink, while equity does not increase enough to offset the reduction. Using IRS tax-return data, Goodman et al. (2023) find no economically significant effect on leverage. All three studies examine the interest cap rather than the rate cut, which is where our model locates the effect on the quantity of debt.

Table 3. The TCJA in the multi-period model

	EBIT volatility 0.28			EBIT volatility 0.56		
Tax regime	Coverage	Debt index	Gain (% of V_U)	Coverage	Debt index	Gain (% of V_U)
Before TCJA: 35%, no cap	2.1	100	23.4	4.0	100	10.9
Rate cut only: 21%	2.1	97	11.1	4.6	87	4.8
Interest cap only: 35% + 30% cap	3.1	68	14.4	5.9	67	7.5
After TCJA: 21% + 30% cap	2.9	72	7.3	5.9	67	3.8

Notes: The first three numerical columns refer to EBIT volatility 0.28 (our firm) and the last three to volatility 0.56. Coverage is mean EBIT over interest at the optimum; optimal debt is indexed to the pre-TCJA level (= 100). Distress cost 20% of V_U, discount rate 10%, lognormal EBIT. All rows are simulated with the same 400,000 draws; because the optimum is flat, differences of a few index points are not meaningful.

6.2 The Singapore tax system and S-REITs

Singapore's corporate tax rate of 17% is one of our sensitivity cases; in the one-period model it moves the debt limit to $D^* = 46.0$, with a maximum gain of about 1.0% of V_U. Singapore is distinctive at the investor level. Dividends from resident companies are exempt under the one-tier system, capital gains are generally untaxed, and individuals are exempt on interest from approved banks and from debt securities (Income Tax Act s13(1); Deloitte, 2025). With $\tau_D = \tau_E = 0$, Miller's T^* equals τ_c, so that, unlike in the United States, the corporate

shield is passed to investors in full. Interest is deductible when the borrowing produces income, there are no thin-capitalisation or EBITDA-based interest caps, and unused losses can be carried forward indefinitely subject to a shareholding test (PwC, 2026; Deloitte, 2025). The frictions of Section 5.2 are therefore weak: a loss year defers the shield rather than eliminating it. In the multi-period model a Singapore firm like ours should borrow up to an interest coverage of 2.2 and gains 8.4% of value (3.6% if EBIT volatility doubles).

Singapore real estate investment trusts (S-REITs) are our real-world attribution test. A REIT that distributes at least 90% of its taxable income is not taxed on the amount distributed, and individual unitholders are generally exempt on the distributions (IRAS, 2025). With almost no tax at either level, our model implies that debt has no tax value, as in Proposition I. S-REITs nevertheless borrow about 40% of their assets (sector average gearing reported by SGX in July 2025; SGX, 2025), so observed leverage on its own is not evidence of a tax benefit; their borrowing must be explained by other considerations, such as funding acquisitions without issuing new units. Regulation looks like a blunt version of our debt limit but is a different constraint. Since 28 November 2024 MAS has capped aggregate leverage at 50% and required an interest coverage ratio of at least 1.5, measured on trailing twelve-month EBITDA (MAS, 2024). That ratio looks backward at reported earnings rather than forward at the worst state that sets D^* in our model, and it protects creditors rather than trading off a tax shield. The April 2020 episode points the same way: when COVID-19 cut rental income, MAS raised the leverage limit from 45% to 50% and deferred to 1 January 2022 a planned minimum interest coverage of 2.5× for REITs borrowing above 45% (MAS, MOF and IRAS, 2020). The regulator relaxed the constraint in the weak state instead of enforcing it, the opposite of our model's hard stop and closer to how SIA was rescued (Section 6.3).

6.3 Calibration to two Singapore-listed firms

Finally, we calibrate the multi-period model to reported cash flows. We collected operating profit and interest expense from the full-year results of Singapore Airlines (SIA, FY2009/10–FY2024/25) and Sheng Siong Group (2013–2024), shown in Figure E1 (Appendix E). For each firm, next year's EBIT is drawn from a normal distribution centred on its average over the last five years, with a standard deviation equal to that of its year-on-year changes. We use changes rather than levels because both series trend over time. Neither measure is ideal: if EBIT is independent across years, changes overstate its dispersion by a factor of about $\sqrt{2}$, and for Sheng Siong they are further inflated by the 2020 jump. For EBIT volatility between S$15 million (the year-on-year measure divided by $\sqrt{2}$) and a conservative S$30 million, Sheng Siong's optimal coverage ranges from 1.4 to 2.0 and the gain from 14% to 9% of unlevered value. All other assumptions are those of the model: a 17% tax rate, default when EBIT does not cover interest, and a distress cost of 20% of value. Table 4 compares the model's prescription with each firm's actual financing.

Table 4. Model prescription and actual financing (S$ million)

	Singapore Airlines	Sheng Siong
Average EBIT, last five years	801	161
Standard deviation of annual change in EBIT	1,356 (1.69× average)	22 (0.13× average)
Loss years in sample	2 of 16	0 of 12
Model: optimal interest coverage	No debt	1.6
Model: gain at the optimum	0%	11.9% of V_U (9–14% across risk measures)
Actual interest coverage, FY2024/25 / 2024	4.3 (FY2020/21: −9.4)	36 (lease interest only)
Actual debt and cash, FY2024/25 / 2024	Debt 12,914 incl. leases; cash 8,257	No bank debt; leases 126; cash 353

Notes: If EBIT is centred on the full-sample average, Sheng Siong's optimal coverage is 2.0 with a gain of 8.5%; SIA's prescription is "no debt" under every centring considered.

Singapore Airlines. Taken literally, the model says SIA should not borrow at all. We read this as a property of our default rule rather than a finding about SIA: EBIT is negative in about 28% of simulated years, and because the model firm holds no cash, even a token amount of debt defaults in each of them. SIA nonetheless carries about S$12.9 billion of debt and lease liabilities. In FY2020/21, the weak state of our model, it made an operating loss of S$2.5 billion against S$0.27 billion of interest but did not default: a Temasek-backed rights issue and convertible bonds raised S$8.8 billion (Singapore Airlines Ltd, 2020, 2021; details in Appendix E). Distress was paid for through dilution rather than bankruptcy, a cost our proportional-distress assumption does not capture. SIA's borrowing rests on features the model omits: aircraft that serve as collateral, large cash reserves and a majority shareholder able to recapitalise the firm.

Sheng Siong. Sheng Siong's EBIT varies little from year to year, so the model would allow it to borrow until interest coverage is about 1.6, adding roughly 12% to firm value. In fact the firm has no bank debt, S$0.35 billion of cash and an interest coverage of 36 on lease interest alone. This is the debt-conservatism puzzle that Graham (2000) documents for large, profitable, low-risk firms. Our model prices taxes and distress only, and omits the value of spare borrowing capacity for expansion and the preferences of controlling shareholders. The model

ranks the two firms as the trade-off theory predicts, giving the stable firm far more debt capacity than the volatile one, but their actual leverage is the reverse; taxes and distress costs alone cannot explain either choice. As with the joint-hypothesis problem of Lecture 3 (p. 25), the gap between prescription and practice tests the firms and our model together: collateral, cash buffers and a shareholder able to recapitalise must be worth more than our omitted terms allow.

7 Conclusion

Our verdict: the corporate income tax makes financing matter, but it does not make debt a free lunch. In the two-state firm, debt of 40 moves 0.73 of value from the tax authority to investors, which vanishes without the tax (attribution), and no investor can copy it with homemade leverage, because only the firm can deduct interest. The money machine stops at the debt the weak year can repay, $D^* = 44.4$, and as long as distress destroys more than about 3.6% of the cash available in default, borrowing just past that limit lowers firm value below V_U (robustness); estimated distress costs of 10–20% (Andrade and Kaplan, 1998) are well above this threshold. What we did not expect is that a higher tax rate makes each unit of debt more valuable but lowers the safe amount of debt, because the tax also comes out of the weak state's cash. In our presentation we called the prize small, about 1–2% of firm value; that is the one-period result, and once debt is maintained the same mechanism is worth about 10% (Section 5.4), close to Graham's (2000) estimate. Unlike a trial of the equal-borrowing-terms assumption, ours shows homemade leverage failing even when investors borrow on the same terms as firms. Table 5 shows that each extension and application changes the size or location of the gain; none makes debt free, and only investor-level taxes can remove the tax advantage of interest altogether.

Our analysis has limitations. Valuation uses risk-neutral probabilities. In the two-state model this affects only the price of distress, but in Section 6.3 we treat historical EBIT distributions as if they were risk-neutral. Distress tends to occur when the whole market is weak, so its risk-neutral probability exceeds its historical frequency (Almeida and Philippon, 2007); we therefore understate the present value of distress costs and overstate optimal debt. In addition, EBIT is drawn independently each year; distress costs are a fixed proportion of value; and the firm holds no cash buffer, so a single bad year triggers default. We also abstract from other determinants of capital structure, including agency costs, asymmetric information and the option value of unused debt capacity, which Section 6.3 suggests are most important for firms at either extreme. Our conclusion would be weakened by evidence that firms with larger interest deductions are not worth more, by distress costs below about 3.6% of default-state cash, or by a marginal investor who is taxed far more heavily on interest than on equity income. For practitioners, the relevant question is less "what is my tax rate?" than "what is the worst year through which the firm must still service its debt?"

Table 5. Summary of results

Analysis	Main finding	Verdict holds?
Parameter sensitivity (Section 4.1)	Debt limit between 37 and 52; gain 1–2% of V_U; debt just past the limit destroys value if distress costs exceed about 3.6% of default-state cash (base case)	Yes; location of the limit is fragile
Continuous cash flows (Section 4.2)	Interior optimum at $D \approx 34$ with 1% default probability; gain 0.85% of V_U	Yes
Market-priced debt (Section 5.1)	Optimum almost unchanged; shareholders rather than lenders bear expected distress costs	Yes
Usability of deductions (Section 5.2)	Effective shield below τ_c for volatile or highly levered firms; a 30% cap binds below coverage of 3.3 in an average year	Yes; adds a second limit
Personal taxes (Section 5.3)	Advantage per unit of interest from −1.7% (US top bracket) to τ_c (Singapore, tax-exempt investors)	Depends on the investor
Multi-period debt (Section 5.4)	Gain rises to about 10% of value; optimal leverage falls with EBIT volatility	Yes; gain is larger
US TCJA (Section 6.1)	Rate cut lowers the value that debt adds by about 40%; an EBIT-based cap reduces optimal debt by up to a third	Yes
Singapore and S-REITs (Section 6.2)	Shield reaches investors in full; S-REITs approximate the no-tax benchmark under a regulatory coverage floor	Yes
Calibration to two firms (Section 6.3)	Volatile airline: no tax case for debt, yet S$12.9 billion borrowed; stable supermarket: about 12% of value forgone	Ranking yes; actual choices differ; SIA's "no debt" corner reflects our no-buffer default rule

Notes: "Verdict" means that corporate tax makes debt valuable up to a limit set by the firm's worst state, and never free.

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Group Contribution Statement and AI-use Disclosure

Member	Responsible part	Core contribution
Lao Xiusheng	Research question and assumptions	Clarified that the main assumption on trial is no corporate tax; distinguished the lecture's Modigliani–Miller assumptions from the group's own model settings.
Xu Riqi	Modigliani–Miller benchmark	Used the two-state example and no-arbitrage reasoning to explain why, without tax, $V_L = V_U$.
Wu Kaluk	Tax shield	Added corporate tax, calculated the tax shield, and explained why debt can make $V_L > V_U$.
Jiang Yumeng	Debt boundary and distress	Increased debt, identified when the weak state can no longer service debt, and showed how distress costs affect firm value.
Li Jiabao	Sensitivity and robustness	Varied the tax rate, distress cost and EBIT distribution to test which conclusions are robust and which are model-specific.
Li Jingwen	Interpretation and limitations	Explained the final economic meaning, treated Modigliani–Miller as a benchmark, and identified model limitations and conditions that could change the conclusion.

All six members reviewed the full analysis, discussed the main result, and prepared jointly for presentation questions.

AI-use disclosure. Generative AI (ChatGPT) was used as a support tool for brainstorming, drafting/editing language, coding assistance, and checking presentation/report structure. All model assumptions, calculations, sources, citations, and final interpretations were reviewed and verified by the group, which remains responsible for the submitted work. The final revision also used Claude (Anthropic) to check the draft against the course materials and the cited sources, rerun and tidy the code, recompute numbers, propose edits and typeset the paper in LaTeX; the group reviewed every change before accepting it.

Appendix A Assumptions

Table A1. Research question and model assumptions

Item	Value	Remark
Research question	Does the corporate tax turn debt into a free lunch?	On trial: no corporate tax; distress costs added only to bound debt
Benchmark	$V_L = V_U$	No taxes, no distress costs, equal borrowing terms (Lecture 1, p. 49)
EBIT, strong / weak state	120 / 60	Two-state firm of Lecture 1 (pp. 9, 47)
State probabilities	0.5 / 0.5	Symmetric states
Promised payment in benchmark example	80	Section 3.1 only
Horizon	One period; liquidation at date 1	Multi-period version in Section 5.4
Discount rate	10%	Risk-free rate; risk-neutral probabilities
Debt at the starting point	40	Base case of the investigation
Borrowing rate	10%, independent of leverage	Market-priced version in Section 5.1
Promised repayment	$44 = 40 \times 1.10$	
Corporate tax rate τ_c	20%	17% and 30% in Section 4
Interest / principal deductible	Yes / No	Source of the shield
Loss carry-forward	None	Discussed in Sections 5.2 and 6.2
Personal taxes / trading costs	None / none	Personal taxes in Section 5.3
Distress cost c	20% of after-tax cash in default (of V_U in Sections 5.4–6)	5% and 40% in Section 4
Continuous cash flow	Lognormal, mean 90, s.d. 25	200,000 draws; s.d. 30 as a check

Notes: All values are illustrative rather than data on a real company, except where Section 6 cites a source.

Appendix B Derivations

Tax shield. Interest $r_D D$ is deducted before tax, so tax falls by $\tau_c r_D D$ in every state with positive taxable income. Received at date 1, the saving is worth $\tau_c r_D D/(1+r)$, which is $0.8/1.1 = 0.727$ for $\tau_c = 20\%$, $r_D = r = 10\%$ and $D = 40$.

Debt capacity. Weak-state after-tax cash is $X_w - \tau_c(X_w - r_D D)$. Equating it to $(1+r_D)D$ gives (4). The same expression with other parameters gives the limits reported in Section 4.

Market-priced debt. For each D, F solves $\mathrm{E}[\min(F, A)$ in solvent states; $(1-c)A$ in default$]/(1+r) = D$, with deductible interest $F - D$; the break-even rate is $F/D - 1$. Under a fixed rate, shareholders' gain from borrowing D is $V_L - V_U + (D - V_{\text{debt}})$, where the second term is the lenders' loss.

Usable shield. With EBIT $\sim \mathrm{N}(\mu, (v\mu)^2)$, interest $I = \mu/k$ (coverage k) and $\mathrm{EBIT}^+ = \max(\mathrm{EBIT}, 0)$, the saving is $\tau_c \min(I, \mathrm{EBIT}^+)$ without carry-forward and $\tau_c \min(I, 0.3\,\mathrm{EBIT}^+)$ under a 30%-of-EBIT cap. We report $\mathrm{E}[\text{saving}]/(\tau_c I)$ from 300,000 draws (150,000 for Figure 3a).

Personal taxes. One unit of interest delivers $1 - \tau_D$ to a lender, whereas the same pre-tax unit paid to shareholders delivers $(1-\tau_c)(1-\tau_E)$; the difference relative to $1 - \tau_D$ gives (5).

Multi-period model. Annual EBIT is i.i.d. and $V_U = (1-\tau_c)\mathrm{E}[\mathrm{EBIT}]/r$. The firm issues perpetual debt D with annual coupon $I = rD$ and defaults in the first year in which after-tax cash does not cover I (equivalently EBIT $< I$ without a cap), which occurs with probability p each year. At default the firm loses cV_U and continues unlevered. Summing the expected shield in solvent years and the expected distress cost over all years gives

$$V_L - V_U = \frac{\mathrm{E}[\text{shield} \times \mathbf{1}\{\text{solvent}\}] - pcV_U}{r + p}, \tag{B1}$$

which reduces to (7) without a cap and to $\tau_c D$ when $p = 0$. Table 3 uses a simulated version, with or without a 30% interest cap; Section 6.3 draws EBIT from a normal rather than a lognormal distribution.

Appendix C Additional results

Table C1. Market-priced debt in the two-state model

D	$V_L - V_U$, fixed rate	Lenders' loss, fixed rate	$V_L - V_U$, market-priced	Break-even rate
40	0.73	0.00	0.73	10.0%
44	0.80	0.00	0.80	10.0%
45	−3.63	4.72	−2.16	29.9%
50	−3.55	7.18	−1.32	37.2%
55	−3.46	9.65	−0.47	43.3%
60	−3.38	12.11	0.38	48.3%

Notes: Beyond the debt limit only the strong state services the debt, so the promised coupon rises sharply; because that coupon is deductible in both states (in the weak state although it is never paid), market pricing moderates the fall in value and at $D = 60$ even makes it a gain. This is an artefact of the two-state model; in the lognormal model (Table C2) the two pricing rules give similar firm values up to $D = 50$.

Table C2. Market-priced debt in the lognormal model

D	$V_L - V_U$, fixed	P(default), fixed	Shareholders' gain, fixed	$V_L - V_U$, market	Break-even rate	P(default), market
20	0.36	0.0%	0.36	0.36	10.0%	0.0%
30	0.53	0.3%	0.55	0.53	10.1%	0.3%
34	0.56	1.0%	0.65	0.57	10.3%	1.0%
35	0.55	1.3%	0.68	0.57	10.4%	1.4%
40	0.43	4.1%	0.89	0.51	11.4%	4.4%
45	0.05	9.6%	1.31	0.19	13.8%	11.5%
50	−0.66	18.0%	2.06	−0.58	19.3%	25.0%
55	−1.68	28.5%	3.29	−2.19	32.5%	49.9%

Notes: 200,000 draws, lognormal EBIT with mean 90 and standard deviation 25.

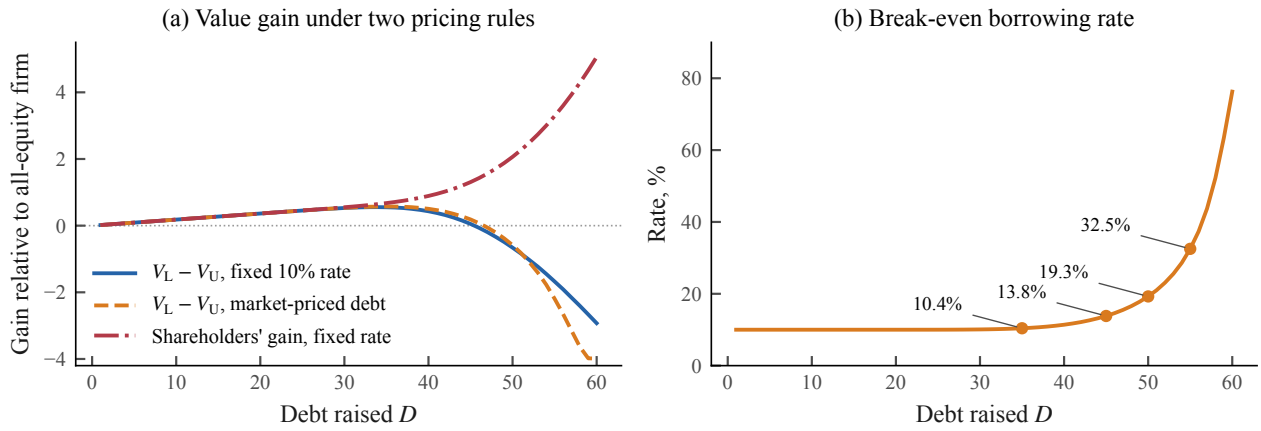


Figure C1. Market-priced debt in the lognormal model. (a) Value gain under fixed-rate and break-even pricing, and the gain perceived by shareholders under a fixed rate. (b) Borrowing rate at which lenders break even.

Table C3. Usable share of the statutory tax shield (%), by interest coverage

	Interest coverage, E[EBIT]/interest						
EBIT volatility	1	1.5	2	3	3.33	5	10
0.28	89	98	99	100	100	100	100
0.28 + cap	30	45	60	84	89	98	100
0.60	77	86	89	92	92	93	94
0.60 + cap	30	45	58	74	77	86	92
1.00	68	74	77	80	80	82	83
1.00 + cap	32	46	56	66	68	74	80

Notes: No loss carry-forward; "+ cap" adds a 30%-of-EBIT limit on deductible interest. Normal EBIT, 300,000 draws.

Table C4. Present value of the shield on $D = 40$ maintained for T years ($\tau_c = 20\%$, $r = 10\%$)

T (years)	1	5	10	30	Perpetual
PV of shield	0.73	3.03	4.92	7.54	8.00

Appendix D Reproducibility

The code, data and LaTeX source of this paper are at https://github.com/jingwen013-design/mm-on-trial (release v1.0, 19 October 2026). The notebook mm_on_trial_report.ipynb follows the order of the paper. By default it loads the cached results of exp1.py (Section 5.1), tables.py (Tables 3 and C1–C3), calib2.py (Section 6.3), extras.py (numbers quoted in the text that no table shows) and the figures from figs.py and fig_firms.py (with style.py), and reruns them all when RECOMPUTE is set. model.py holds the model and firms.json the firm data. Two-state results and the multi-period results of Section 5.4 are exact; the other lognormal and normal results use fixed random seeds with 150,000–400,000 draws, so that reruns reproduce every reported figure. Market-priced debt is solved by root-finding on the lenders' break-even condition. As an independent check, independent_check.py, which does not use model.py, values the lognormal case by numerical integration and reproduces $V_U = 65.45$, $D^* = 44.44$, Table 1 and the lognormal optimum ($D \approx 33.8$, gain 0.557, default probability 0.92%).

Appendix E Firm-level detail

Singapore Airlines' 2020–21 financing. The S$8.8 billion raised in 2020 combined a rights issue priced 31.8% below the theoretical ex-rights price with mandatory convertible bonds, backstopped by Temasek; SIA also arranged a S$4 billion bridge loan and in 2021 issued a further S$6.2 billion of mandatory convertible bonds (Singapore Airlines Ltd, 2020, 2021).

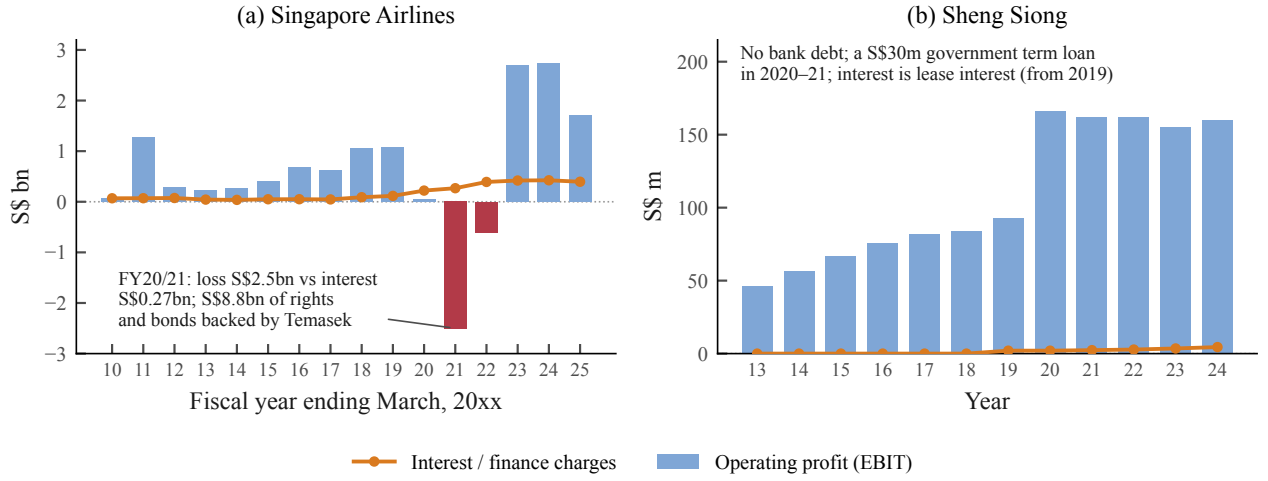


Figure E1. Operating profit (bars) and interest expense (line), S$. Loss years are shown in red. SIA's interest includes lease interest from FY2019/20, when IFRS 16 brought leases onto the balance sheet; Sheng Siong's interest is lease interest only. Sources: Singapore Airlines Ltd (2010–2025); Sheng Siong Group Ltd (2015–2025).